The Augmented Synthetic Historical Control Method

Introduction

The synthetic control method (SCM) by Abadie et al. [2010] is one of the most popular methods in panel data econometrics for treatment effect estimation, second only to difference-in-differences. The idea of an SCM is very simple: it simply posits that the treated unit's untreated potential outcomes may be understood as some function/weighted average of a set of donor pool units, or unexposed/untreated entities. Many extensions have been proposed to this end, including singular value decomposition [Amjad et al., 2018, Costa et al., 2023], penalized regression models [Abadie and L'Hour, 2021], Bayesian methods [Fernández-Morales et al., 2025, Kim et al., 2020], and other extensions [Abadie, 2021].

However, as with difference-in-differences, SCMs very generally rely on the availability of a donor pool with which to compare the evolution of the treated outcome to. This may not always holds in practice, particularly with global events like pandemics, financial crises, or in other instances where data for suitable donor units are either impossible to collect or do not exist in principle. Chen et al. [2024] develop the synthetic historical control method (SHC). Instead of untreated donors (like other cites or schools or states), SHC uses the pre-treatment time series of the treated unit as its own donor pool, comparing an evaluation segment to other time periods in the treated unit's history by breaking the pre-treatment periods into segments. The basic principle of the method is that the future will look like the past, all else equal. Chen et al. [2024] use quadratic programming to solve for the weights for the donor segments, using a weighted average of them to predict the post-treatment counterfactual, using COVID-19 and Brexit as empirical cases.

However, Chen et al. [2024] use convex weights to estimate the counterfactual. This means that the model's predictions are restricted only to time segments that we have observed before, and may not be suited for complex time-series data where noise or non-smooth dynamics make a pure convex average inadequate. In such cases, exact pre-treatment fit may be impossible, a challenge that also arises in standard SCM applications [Abadie et al., 2010].

This paper introduces the Augmented Synthetic Historical Control Method (ASHC), which relaxes the strict convexity requirement by allowing limited extrapolation away from the SHC solution. ASHC adds a quadratic penalty on deviations from the SHC weights, together with a stabilizing penalty in directions of low variance. This balances the desire for close fit with the need for stability, ensuring that extrapolation improves flexibility without introducing uncontrolled overfitting.

From a theoretical standpoint, ASHC departs from the traditional linear factor model framework used to justify SCM. In SCM, identification is motivated by assuming that untreated potential outcomes follow a factor structure across units. By contrast, SHC and ASHC operate in a purely time-series context: the donor pool consists of overlapping pre-treatment segments of the treated unit itself. Our assumptions therefore follow the semiparametric regression tradition for time-series data. Specifically, we assume that the latent trend is smooth enough to be well-approximated by a truncated Taylor expansion, that noise is bounded (or sub-Gaussian), and that the donor matrix has bounded operator norm. These conditions, while weaker than a full parametric data-generating process, are sufficient to establish finite-sample deviation bounds, asymptotic consistency, and convergence rates for the ASHC estimator.

In this sense, ASHC provides a bridge between SCM-style identification and nonparametric time-series methods: it retains the interpretability of synthetic control weights while relaxing the convexity constraint to better handle noisy or complex dynamics.

References

Alberto Abadie. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2):391–425, 06 2021. URL 10.1257/jel.20191450.

Alberto Abadie and Jérémy L'Hour. A penalized synthetic control estimator for disaggregated data. *Journal of the American Statistical Association*, 116(536):1817–1834, 2021. doi: https://doi.org/10.1080/01621459.2021.1971535.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010. URL https://doi.org/10.1198/jasa.2009.ap08746.

Muhammad Amjad, Devavrat Shah, and Dennis Shen. Robust synthetic control. *The Journal of Machine Learning Research*, 19(1):802–852, 2018.

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.

Luis Costa, Vivek F. Farias, Patricio Foncea, Jingyuan (Donna) Gan, Ayush Garg, Ivo Rosa Montenegro, Kumarjit Pathak, Tianyi Peng, and Dusan Popovic. Generalized synthetic control for testops at abi: Models, algorithms, and infrastructure. *INFORMS Journal on Applied Analytics*, 53(5):336–349, 2023. URL https://doi.org/10.1287/inte.2023.0028.

Esteban Fernández-Morales, Arman Oganisian, and Youjin Lee. Bayesian shrinkage priors for penalized synthetic control estimators in the presence of spillovers, 2025. URL https://arxiv.org/abs/2501.08231.

Sungjin Kim, Clarence Lee, and Sachin Gupta. Bayesian synthetic control methods. *Journal of Marketing Research*, 57(5):831–852, 2020. doi: 10.1177/0022243720936230.